

MARITIME SYSTEMS LAB REPORT

on

ACCELERATION

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Date of Submission:

16/02/2026

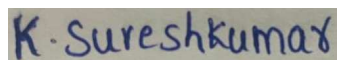


MASTER OF SCIENCE IN EMBEDDED SYSTEM DESIGN

Winter Semester – 2025/26

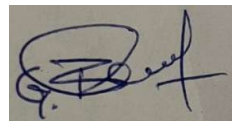
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ABSTRACT

In this laboratory work, an acceleration measurement system was designed, assembled, and tested. The assembled measuring circuit was used to record elevator motion data in the T-building of Hochschule Bremerhaven during travel from the second floor to the third floor. The recorded data were calibrated and from the measured acceleration, the velocity and travelled distance were calculated, enabling the estimation of the floor height, which was determined to be approximately **3.94 m**.

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1. Introduction

In this laboratory work, an acceleration measurement system was designed, assembled, and tested using the ADXL335 acceleration sensor from Analog Devices [1]. The system was used to record acceleration data during elevator motion in the T-building of Hochschule Bremerhaven. The measured data were calibrated and processed to calculate velocity and travelled distance. This work demonstrates the complete process from acceleration measurement to data analysis in a practical application.

2. Task description

The tasks of this laboratory work are as follows:

- i. To assemble and set up the acceleration measurement circuit, including the acceleration sensor.
- ii. To test the circuit operation and determine the cut-off frequency using the frequency response method.
- iii. To determine the cut-off frequency of the system using the step response method and to compare the results with the frequency response method.
- iv. To acquire acceleration data using a microcontroller-based data acquisition system.
- v. To record acceleration measurements during elevator motion in the T-building of Hochschule Bremerhaven between the second and third floors.
- vi. To calibrate the measured acceleration data and process it to calculate velocity, travelled distance, and estimate the floor height.

3. Theoretical background

3.1 Acceleration measurement

Acceleration measurement is used to analyse motion and to derive related quantities such as velocity and displacement. In this laboratory work, acceleration is measured directly using an ADXL335 micro electrical mechanical system (MEMS) acceleration sensor [1], while velocity and distance are obtained indirectly by processing the measured data.

Acceleration is defined as the rate of change of velocity with respect to time and is given by:

$$a = \frac{dv}{dt} \quad (1)$$

Where a is acceleration, v is velocity and t is time.

From (1) we can observe that velocity can be obtained by integrating the acceleration over time:

$$v = \int_0^t a. dt + v_0 \quad (2)$$

Where v_0 is initial velocity, which will be zero in our case.

Velocity is defined as rate of change of distance with respect to time and is given by:

$$v = \frac{ds}{dt} \quad (3)$$

Where s is distance.

From (3) we can observe that distance is obtained by integrating velocity over time:

$$s = \int_0^t v. dt + s_0 \quad (4)$$

Where s_0 is initial distance, which will be zero in our case.

3.2 Acceleration sensor

The acceleration sensor, which we use, is the integrated component ADXL335 from Analog Devices [1]. It is based on a micro electric mechanical system (MEMS) with a mass pendulum. When the pendulum is deflected capacities are internally modified and, thus, the output voltage is affected. As the relative deflection of the pendulum is dependent on the acceleration of the sensor, the level of the output voltage becomes the measure of acceleration. The relation between these two parameters is linear. The static acceleration of gravity as well as dynamic acceleration, provoked by movement, can be measured. The ADXL335[1] measures acceleration in three axes.

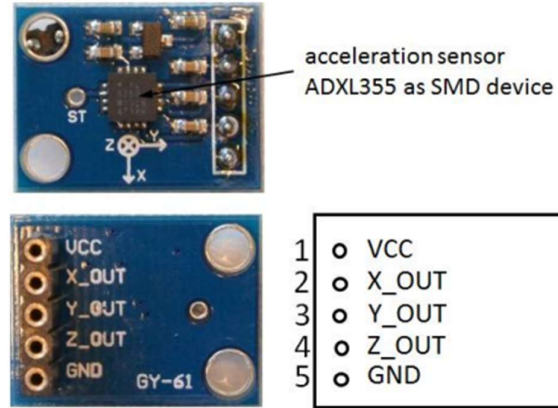


Figure 1: Sensor module GY-61 for measurement of acceleration [2]

In Figure 1, VCC is the power supply (1.8V to 5V), X_OUT, Y_OUT and Z_OUT are x channel, y channel and z channel analog outputs respectively and GND is ground. We know that ADXL335[1] measures acceleration in three axes but for our experiment the elevator moves only in one direction, therefore we have used only one channel output and that is X_OUT.

In operating state, the output voltage adjusts according to following equation

$$V_S = V_{off} + (g + a_d) \cdot S \quad (5)$$

Where,

- V_S is sensor output voltage
- V_{off} is offset voltage (output voltage without acceleration- Zero g Bias level)
- g is acceleration due to gravity ($=9.81 \text{ m/s}^2$)
- a_d is dynamic acceleration
- S is Sensitivity

From the datasheet of ADXL335 from Analog Devices [1],

- Sensitivity $S = 300 \text{ mV/g}$
- Offset voltage $V_{off} = 1.5 \text{ V}$

4. Experimental setup

4.1 Circuit assembly:

Figure 2 illustrates the test circuit for the acceleration measurement system. The resistor values were determined prior to assembly, and the circuit was then built on a laboratory experimental PCB for acceleration measurements.

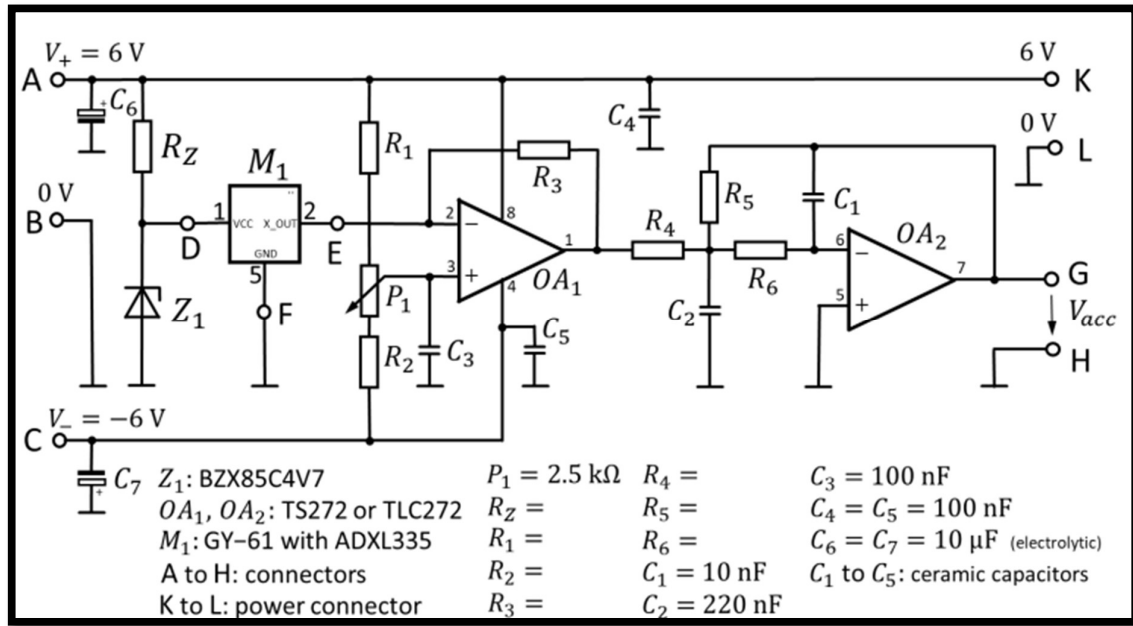


Figure 2: Schematics of the test circuit with sensor module and operational amplifiers. [2]

The schematic can be divided into four main functional blocks:

- Power supply section
- Sensor module
- First amplification stage
- Lowpass filtering stage.

4.1.1 Power supply section:

The acceleration sensor module M_1 is supplied with the required voltage using a Zener diode. When a diode is operated in reverse bias and the applied voltage exceeds the breakdown voltage, the reverse current increases sharply. For standard diodes, this effect is generally undesirable. However, in Zener diodes, the breakdown voltage is precisely defined and is referred to as the Zener voltage V_Z .

Zener diodes are commonly used for voltage stabilization in direct current circuits. When operated in the breakdown region, they maintain an approximately constant voltage across their terminals despite variations in current.

In this experiment, the Zener diode BZX85C4V7 is used, which has a specified Zener voltage of $V_Z = 4.7$ V. According to the datasheet [3], this voltage remains relatively stable at a Zener current of approximately $I_Z = 40$ mA. You can observe the following in Figure 3. Zener diodes are often abbreviated as Z-diodes.

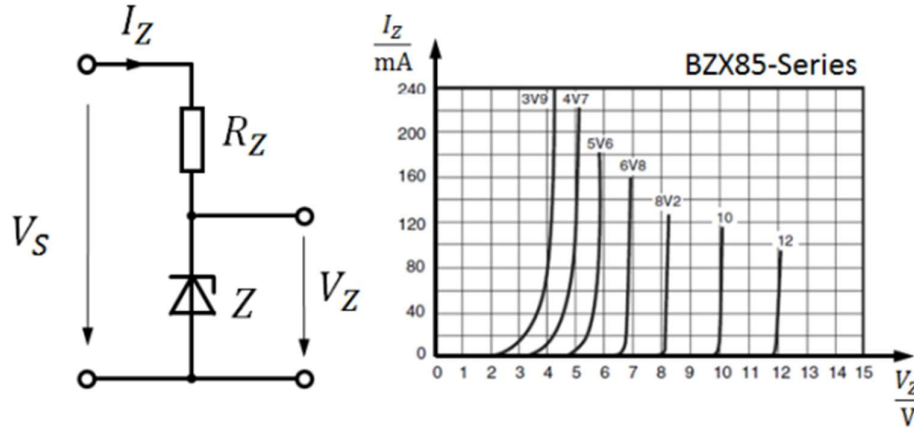


Figure 3: Z-diode to stabilize the voltage with breakthrough voltage and current. [2]

4.1.2 Sensor module

The positive terminal of the Zener voltage V_Z is connected to the VCC input of the sensor module, and the negative terminal is connected to ground (GND). The sensor provides an analog output voltage proportional to the acceleration in the X-direction. The remaining two axes are not used in this experiment. The ground connection is shared for both power supply and signal reference.

4.1.3 First amplification stage

The first operational amplifier OA_1 is configured as a subtracting amplifier to compensate for the sensor offset voltage. The second operational amplifier OA_2 is implemented as an inverting amplifier combined with an active second-order low-pass filter. A dual operational amplifier (TS272) [4] is used, which integrates both amplifiers in a single package. Decoupling capacitors are placed close to the supply pins to ensure stable operation. The circuit is powered using both positive and negative supply voltages.

The amplification factor is selected such that the sensor output voltage is scaled to the required output range of the test circuit. The offset voltage is adjusted using a voltage divider with a trimming potentiometer P_1 . A blocking capacitor is included to prevent unwanted oscillations.

4.1.4 Low Pass Filtering Stage:

The second amplifier stage forms an active second-order low-pass filter with multiple negative feedback. This filter suppresses high-frequency components of the acceleration signal, which is necessary before analog-to-digital conversion.

The conditioned output signal is then connected to the microcontroller-based data acquisition board. The connectors *K* and *L* provide the required power supply for the microcontroller during measurements.

$$A = \frac{(V_{acc,max} - V_{acc,min})}{(V_{S,max} - V_{S,min})} = \frac{5\text{ V} - 0\text{ V}}{1.95\text{ V} - 1.35\text{ V}} = 8.33$$

Where,

- $V_{acc,max}$ is maximum output voltage we expect between G & H
- $V_{acc,min}$ is minimum output voltage we expect between G & H
- $V_{S,max}$ is maximum sensor output voltage
- $V_{S,min}$ is minimum sensor output voltage

We get $V_{S,max}$ and $V_{S,min}$, by solving equation (5) using the values from the datasheet of ADXL335 from Analog Devices [1].

4.2 Determination of component values

The component values used in Figure 2 are summarized in Table 1. The resistor values were chosen in accordance with the E12 standard series.

Table 1: Values of components used in Figure 2

No	Component	Value (in K Ω)
1	Resistor, R_z	0.033 K Ω
2	Resistor, R_1	12 K Ω
3	Resistor, R_2	15 K Ω
4	Resistor, R_3	56 K Ω
5	Resistor, R_4	68 K Ω
6	Resistor, R_5	330 K Ω
7	Resistor, R_6	330 K Ω
8	Potentiometer, P_1	2.5 K Ω

9	Capacitor, C_1	10 nF
11	Capacitor, C_3	100 nF
12	Capacitor, C_4	100 nF
13	Capacitor, C_5	100 nF
14	Capacitor, C_6	10 μ F
15	Capacitor, C_7	10 μ F

5. Measured Data

5.1 Functional Testing of the Assembled Circuit

All required components for the circuit shown in Figure 2 were collected according to the component values listed in Table 1. The components were placed on the experimental printed circuit board in a structured manner to ensure proper spacing and ease of soldering as shown in Figure 4.

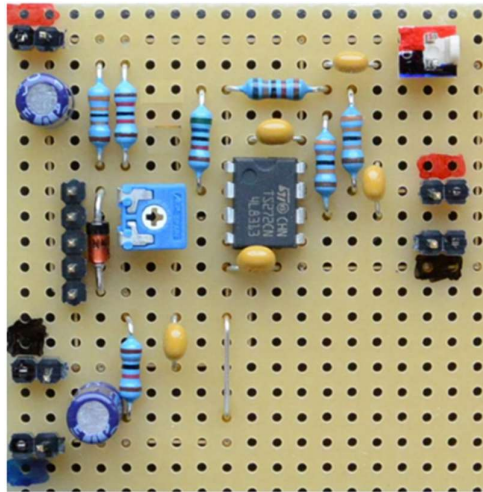


Figure 4: Positioning of the components on the experimental printed circuit board. [2]

For the initial setup and verification, the sensor module was not connected to the circuit. The circuit was powered using a laboratory power supply providing $V_+ = 6\text{ V}$ and $V_- = -6\text{ V}$. To simulate the sensor output voltage V_S , a laboratory power supply was used together with a voltage divider network, as shown in Figure 5. The internal sensor resistance R_S was simulated using an external resistor which is $32\text{ K}\Omega$, from the datasheet [1].

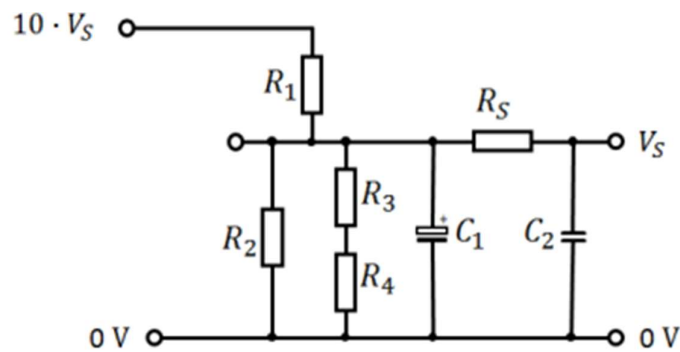


Figure 5: Voltage divider 10:1 with simulated internal resistor R_S of the acceleration sensor [2]

The output voltage V_{acc} was measured by applying $V_{S,min}$ and $V_{S,max}$ which are calculated using equation (5). Here we use 10 times $V_{S,min}$ and 10 times $V_{S,max}$ since it is 10:1 voltage driver and check whether we are getting desired output voltage V_{acc} which are listed in Table 2. The trimmer P_1 was adjusted to achieve the desired offset compensation.

Table 2: Input and Expected Output Voltages for verification

No	Applied input voltage (in Volts)	Expected output voltage (in Volts)
1	13.5	0
2	19.5	5

5.2 Measuring Cutoff-frequency

To measure the cutoff-frequency we have two methods:

- Frequency response
- Step function response

5.2.1 Frequency response:

The cut-off frequency is that frequency, where the relation between output voltage and input voltage has decreased to $\frac{1}{\sqrt{2}}$ with reference to the maximal value. A sine-wave signal is generated with a peak-to-peak voltage of $V_{pp} = 1$ V and an offset voltage of $V_{off} = 1.5$ V in oscilloscope. The frequency of the input signal is varied within the range of 1 Hz to 50 Hz. The generated AC signal is first applied to the amplifier stage, where the signal is amplified and the amplified signal then passes through the low-pass filter, which suppresses high-frequency components. The final output voltage is measured using the oscilloscope. For each input frequency, the peak-to-peak output voltage is recorded (these values are noted in Table 3) to plot the frequency response (Figure 6).

Table 3: Oscilloscope measurements

Frequency (in Hz)	Peak -Peak Output voltage (in Volts)
1	7.2
2	7.1
3	7.0
4	6.8
5	6.7
6	6.4
7	6.2
8	5.8
9	5.4

Frequency (in Hz)	Peak -Peak Output voltage (in Volts)
10	4.9
12	4.1
14	3.4
16	2.8
18	2.3
20	2.0
22	1.7
24	1.4
26	1.2
28	1.0
30	1.0
35	0.7
40	0.6
45	0.6
50	0.4

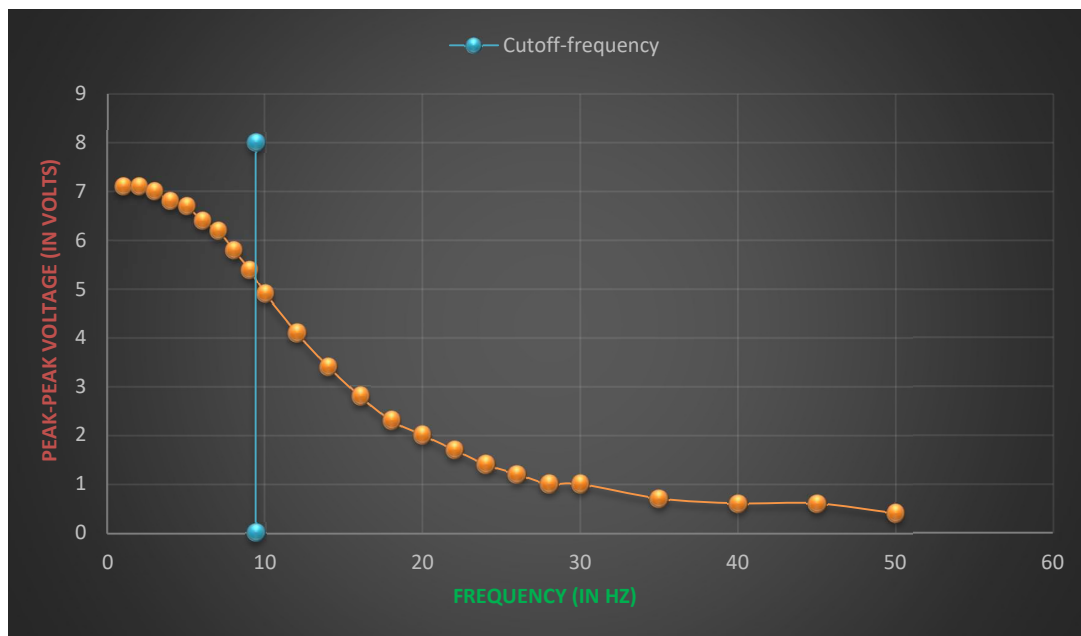


Figure 6: Frequency response plot

Based on the frequency response shown in Figure 6, the cut-off frequency is determined as $f_c = 9.4$ Hz.

5.2.2 Step function response:

The cut-off frequency of the test circuit may also be determined using its step response. A square-wave signal is generated in the oscilloscope with a period of $T = 1$ s. The maximum input voltage is set to $V_{in} = 2$ V and the minimum input voltage to $V_{in} = 1$ V. The time response of the circuit to the step input is shown below in Figure 7.

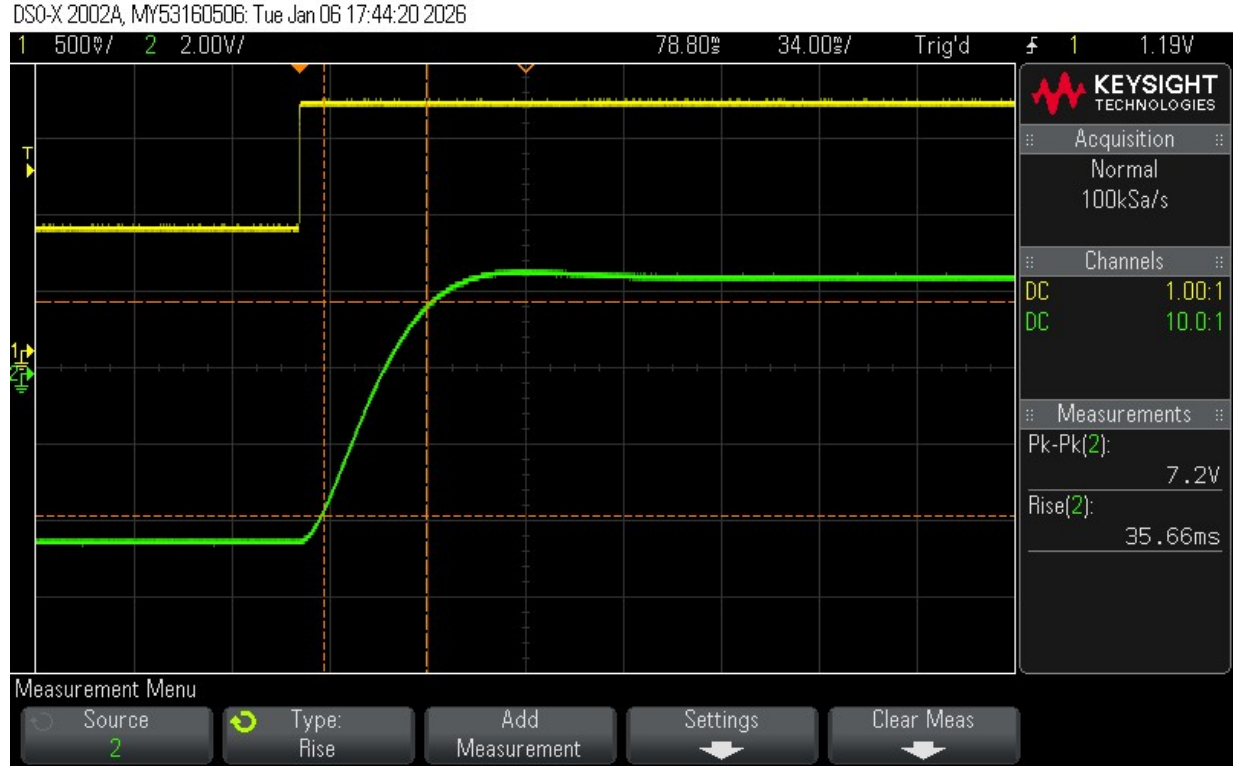


Figure 7: Oscilloscope step response

The slew rate t_s is determined as the time required for the output voltage to change from 10% to 90% of its total amplitude. Based on the measured slew rate, the cut-off frequency is calculated using formula:

$$f_c \approx \frac{1}{3 \cdot t_s} \quad (6)$$

From Figure 6, it is clear that $t_s = 35.66$ ms. Substituting it in Equation (6) gives

$$f_c \approx \frac{1}{3 \cdot (35.66 \cdot 10^{-3} \text{sec})} \approx 9.35 \text{Hz}$$

The cut-off frequency obtained from the step response method closely matches the value derived from the frequency response method in Chapter 5.2.1. The small difference is mainly due to measurement and approximation errors, confirming the correct operation of the circuit.

5.3 Data acquisition

The data acquisition system is based on the ATmega32[5] microcontroller from Microchip. The purpose of this stage is to sample the analog output voltage of the test circuit and transmit the acceleration data digitally via the serial interface. The circuit diagram of the experimental microcontroller board is shown in Figure 8. It should be noted that the diagram is only partially illustrated, and only the components relevant to this experiment are included.

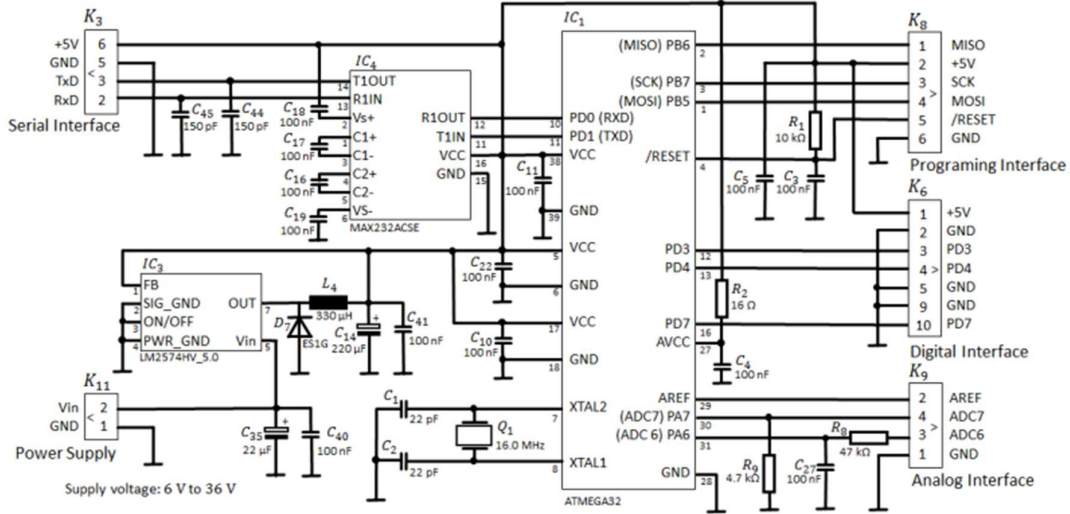


Figure 8: Experimental microcontroller board [2]

5.3.1 Nyquist criterion:

According to the Nyquist–Shannon sampling theorem, a continuous-time signal can be reconstructed accurately if the sampling frequency is at least twice the highest frequency component of the signal. Since the cut-off frequency of the low-pass filter is approximately $f_c \approx 10$ Hz, a sampling frequency of $f_s = 100$ Hz was selected. This value satisfies the Nyquist criterion and additionally provides oversampling, which reduces aliasing effects and improves measurement accuracy. The recorded acceleration data are aggregated and transmitted at a reduced rate of 10Hz. Thus, ten consecutive samples are combined before transmission.

5.3.2 Timer configuration:

To ensure precise and constant sampling, Timer0 of the ATmega32 was configured in Clear Timer on Compare Match (CTC) mode. This configuration is achieved by setting the register bits WGM01 = 1, WGM00 = 0. The OCR0 (Output compare register) value is calculated using

$$OCR0 = \frac{\text{Clock frequency}}{(\text{prescaler} \cdot \text{sampling frequency})} - 1 \quad (7)$$

Where clock frequency is 16 MHz, prescaler is 1024 and sampling frequency is 100Hz. Substituting these values in Equation (7) gives OCRO value as 155.

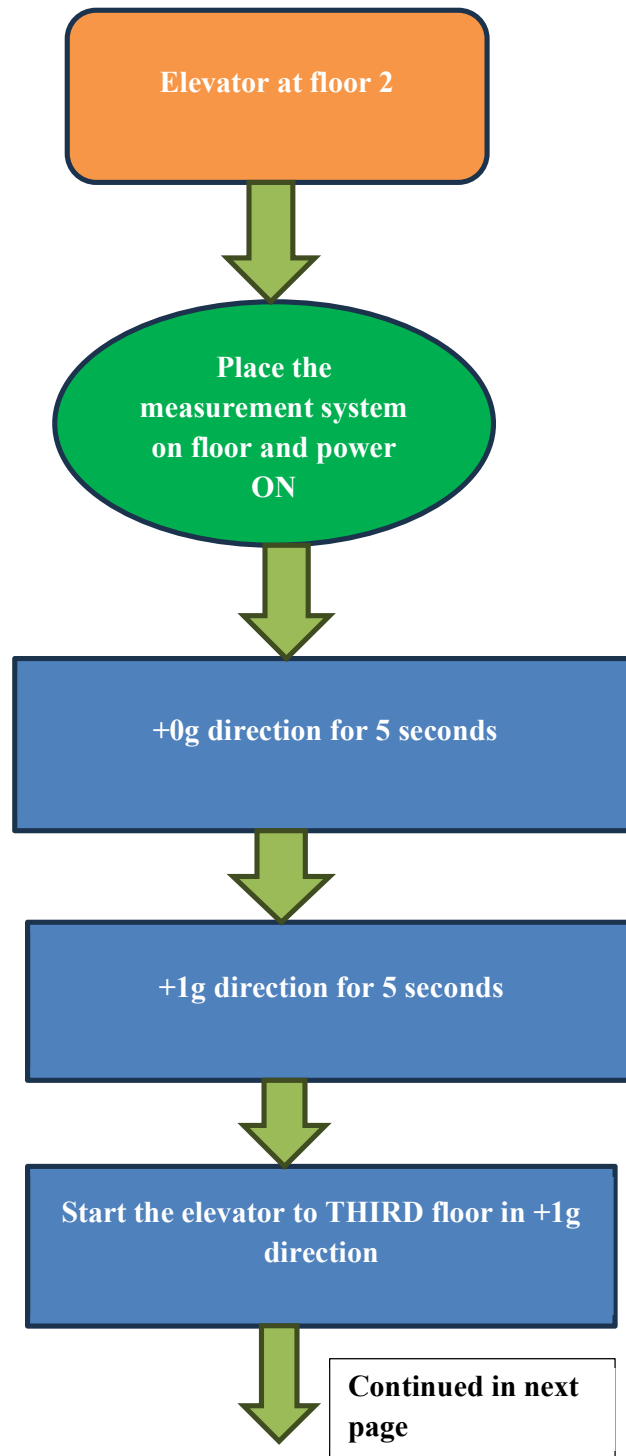
5.3.3 Analog to digital conversion and transmission:

The analog output voltage V_{acc} of the measurement circuit is connected to ADC channel 6 of the ATmega32[5]. At each timer interrupt, a new ADC value is recorded. The measured digital value d_{acc} is proportional to the output voltage of the test circuit and therefore proportional to the acceleration. Ten successive measurements are combined and transmitted as ASCII data via the serial interface at a rate of 10 Hz. This ensures stable data logging.

6. Elevator measurements and height estimation

6.1 Elevator measurements procedure

Figure 9 represents the flowchart of the steps involved in recording the elevator measurements.



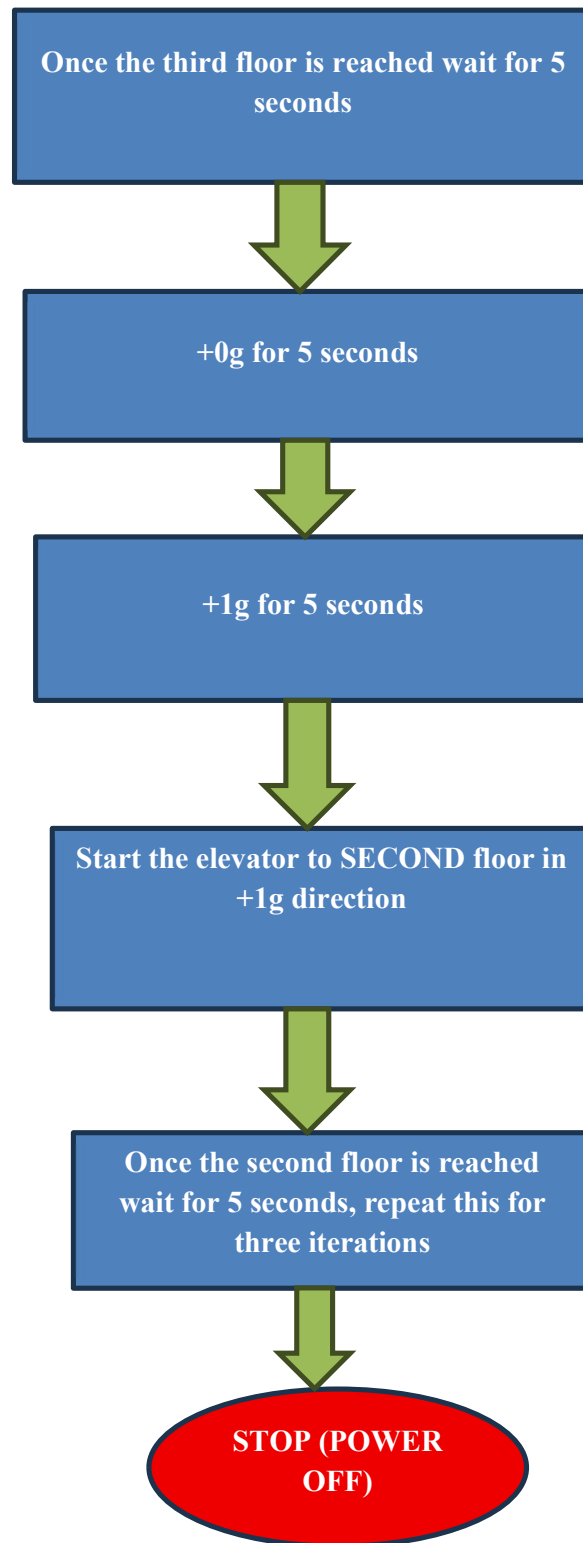


Figure 9: Flowchart of elevator measurements procedure

The data collected after following the procedure outlined in the flowchart (Figure 9) are plotted in Figure 10.

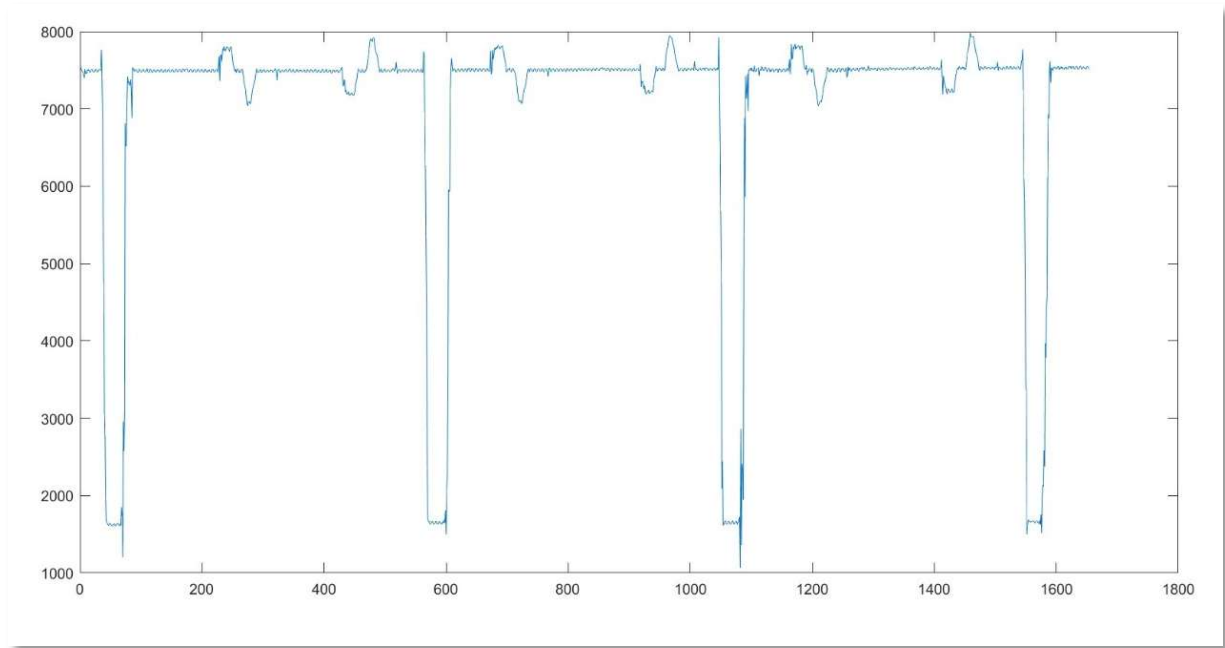


Figure 10: Plot of recorded data

For detailed explanation (explained in Table 4), a single iteration from the above three iterations is illustrated in Figure 11.

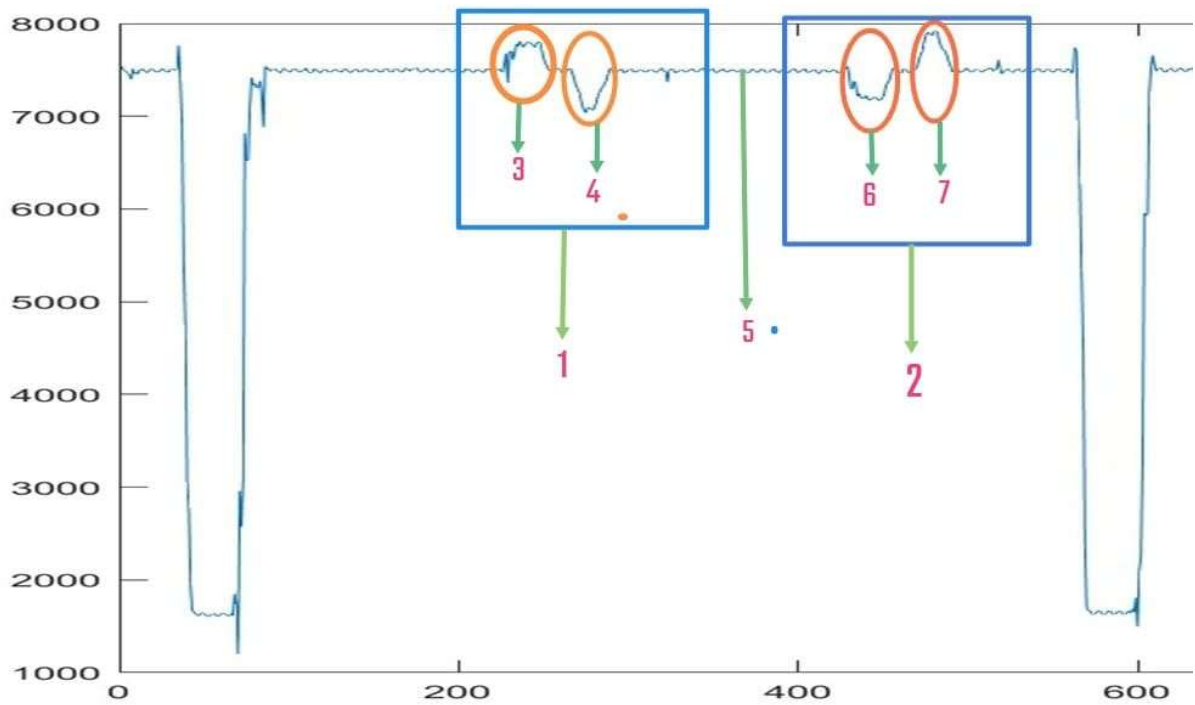


Figure 11: First iteration

Explanation:

Table 4: Explanation of plot of recorded data for first iteration

PHASE	DESCRIPTION
1	Elevator movement from floor 2 to floor 3 (Upwards)
2	Elevator movement from floor 3 to floor 2 (Downwards)
3	This region represents upward acceleration occurring at the moment the elevator initiates its motion from Floor 2 to Floor 3.
4	The spike observed in the opposite direction of region 3 represents deceleration as the elevator approaches to stop at Floor 3.
5	This region indicates that the elevator is stationary at Floor 3.
6	This region indicates downward acceleration, which occurs when the elevator starts moving from Floor 3 to Floor 2.
7	The spike observed in the opposite direction of region 6 represents deceleration as the elevator approaches to stop at Floor 2.

6.2 Calculation of floor height from elevator data:

The acceleration can be calculated using the following equation:

$$a = k_1 \cdot d_{acc} + k_0 \quad (9)$$

Where,

- a is acceleration
- k_1 is scaling factor or gain
- d_{acc} is dynamic acceleration recorded by sensor
- k_0 is offset factor

As we have two known values for every trip in elevator (+0g and +1g), Equation (9) can be further written as

$$1g = k_1 \cdot d_{acc(1g)} + k_0 \quad (10)$$

$$0g = k_1 \cdot d_{acc(0g)} + k_0 \quad (11)$$

Subtracting Equations (10) and (11) gives

$$k_1 = \frac{1g}{d_{acc(1g)} - d_{acc(0g)}} \quad (12)$$

Equation (11) can be written as

$$k_0 = -k_1 \cdot d_{acc(0g)} \quad (13)$$

So, to find k_1, k_0 we need to have $d_{acc(0g)}, d_{acc(1g)}$ which can be obtained by taking average of calibration values from the graph in Figure 10.

From Figure 10, the calibration values for 1g are as follows:

- First iteration: 7509
- Second iteration: 7519
- Third iteration: 7489

$$\text{Therefore, } d_{acc(1g)} = \frac{(7509+7519+7489)}{3} = 7505.66$$

the calibration values for 0g are as follows:

- First iteration: 1667
- Second iteration: 1669
- Third iteration: 1617

$$\text{Therefore, } d_{acc(0g)} = \frac{(1667+1669+1617)}{3} = 1651$$

Substituting these values in Equations (12) and (13) gives $k_1 = 0.00017g$ and $k_0 = -0.28g$

Substituting k_1 and k_0 in Equation (9) gives

$$a = (0.00017 \cdot d_{acc} - 0.28) \cdot g \quad (14)$$

Using Equation (14) and the acceleration readings (d_{acc}) recorded during the elevator trips, the acceleration graph for the first iteration was plotted, as shown in Figure 12.

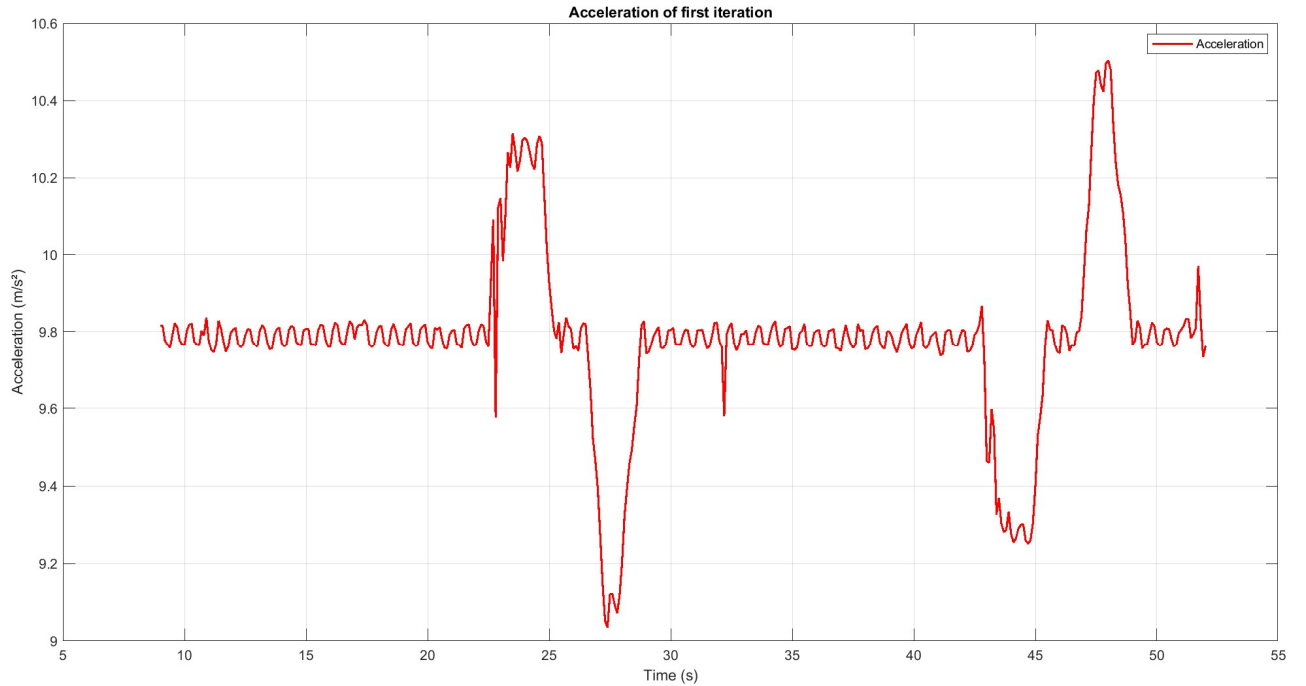


Figure 12: Acceleration plot for first iteration

Figure 12 shows the acceleration plot for the first iteration, which includes both the upward and downward movements of the elevator. However, to determine the height between two floors, data from only one direction of motion is sufficient. It can also be observed that the graph contains noise and disturbances. These can be reduced by applying a low-pass Butterworth filter, and the offset can be corrected accordingly. After performing these modifications, the filtered and offset-corrected acceleration for the upward movement of the first iteration is plotted in Figure 13.

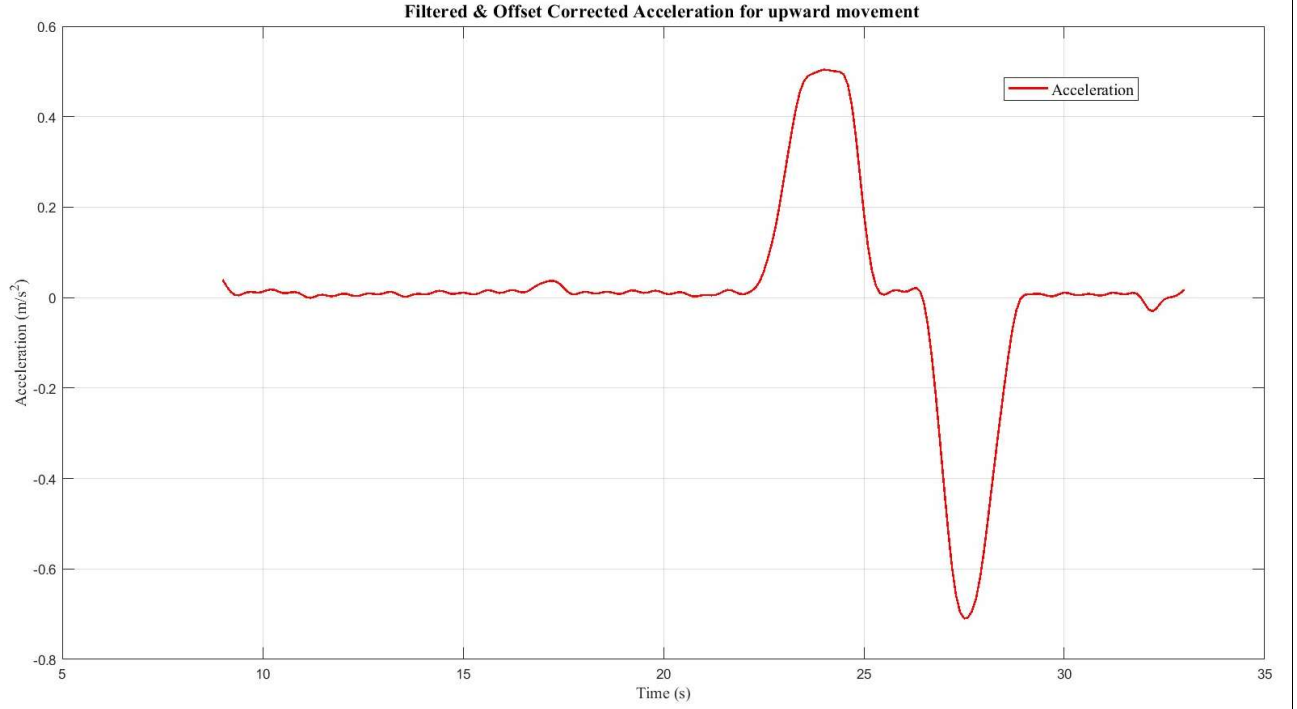


Figure 13: Filtered and offset corrected acceleration for upward movement of first iteration

Velocity:

To plot the velocity, Equation (2) from chapter 3.1 can be used based on theoretical understanding. However, this continuous-time integral cannot be implemented directly in software because we do not have infinitely small-time steps. In practice, the acceleration is recorded as discrete samples, so the integral must be approximated by a summation. Therefore, for programming the velocity calculation, Equation (2) is converted into its discrete form and implemented using Equation (15) and the resulting velocity plot is shown in Figure 14.

$$v_i = \sum_{j=0}^i a_j \cdot \Delta t + v_0 \quad (15)$$

Where,

- v_i is velocity at the instance i
- a_j is acceleration at instance j
- Δt is time instance between two measurements

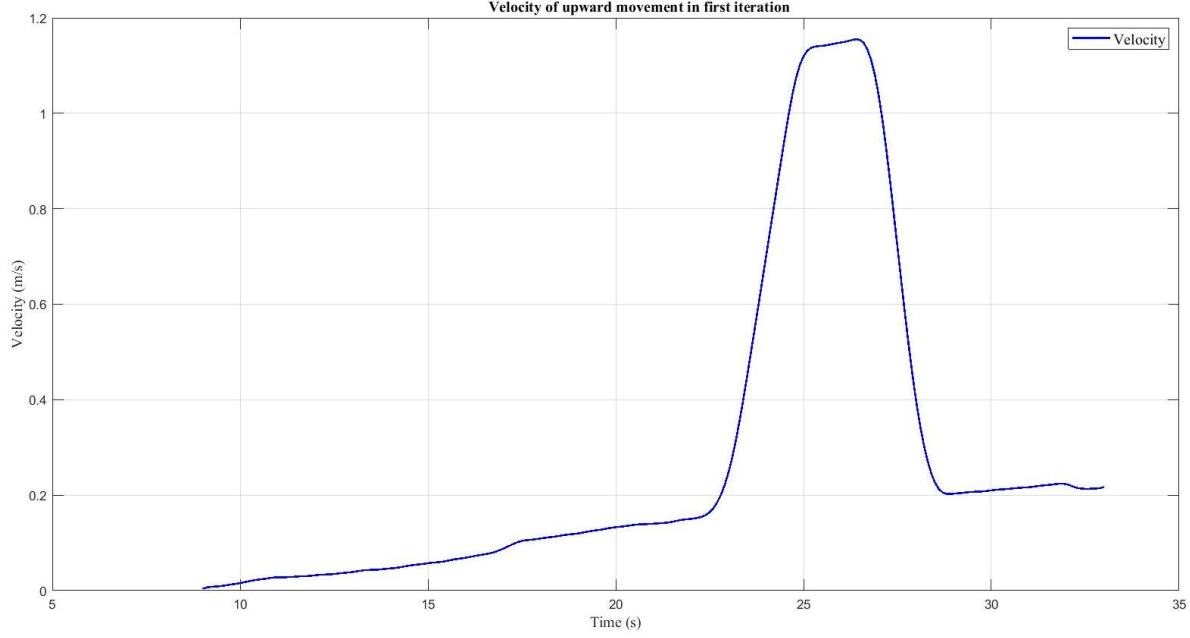


Figure 14: Velocity of upward movement in first iteration

Velocity correction:

The velocity plot shown in Figure 14 is not as expected. Ideally, when the elevator stops, the velocity should return to zero; however, in Figure 14 a small residual velocity remains. This indicates a linearly increasing drift, mainly caused by a small calibration (bias) error in the acceleration signal. Since perfect calibration is difficult to achieve, a correction step is required. Therefore, the linear drift is estimated and subtracted from the velocity signal.

The corrected velocity is obtained as:

$$v_{corrected} = v_{original} - (slope \cdot time) \quad (16)$$

Where,

- $v_{corrected}$ is corrected velocity
- $v_{original}$ is original velocity

$$slope = \frac{(v_{end} - v_{start})}{T} \quad (17)$$

- v_{end} is velocity at last time instant
- v_{start} is velocity at first time instant

This correction forces the velocity to approach zero when the elevator is stationary, as shown in Figure 15.

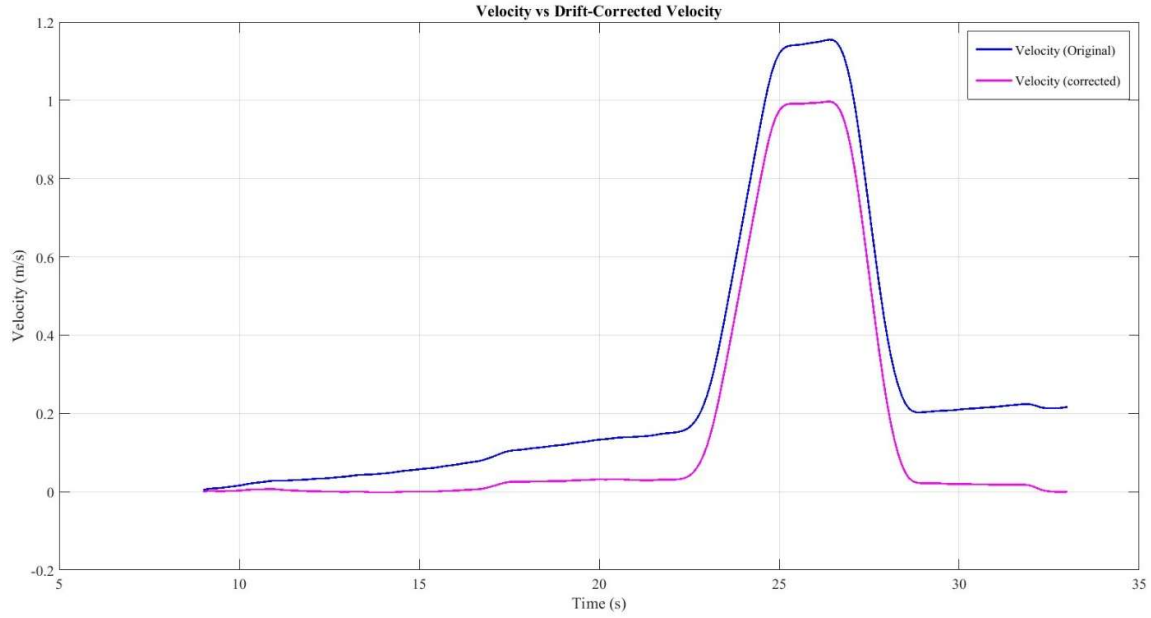


Figure 15: Velocity vs corrected velocity plot

Distance (Height):

Similarly, to plot the distance, Equation (4) from Chapter 3.1 can be used based on theoretical understanding. However, this continuous-time integral cannot be implemented directly in software because we do not have infinitely small-time steps. Therefore, the integral must be approximated by a summation. For programming the distance calculation, Equation (4) is converted into its discrete form and implemented using Equation (18), and the resulting distance plot is shown in Figure 16.

$$s_i = \sum_{j=0}^i v_j \cdot \Delta t + s_0 \quad (18)$$

Where,

- s_i is distance at the instance i
- v_j is velocity at instance j
- Δt is time instance between two measurements
- s_0 is initial distance (which will be zero)

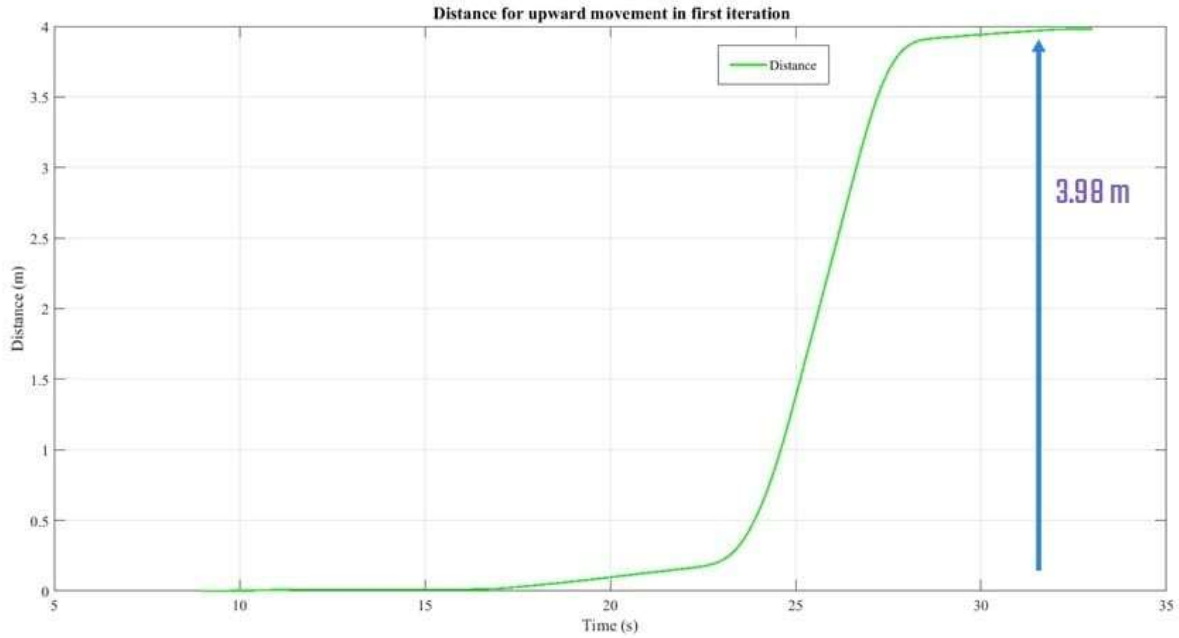


Figure 16: Distance of upward movement in first iteration

From Figure 16, it can be observed that the calculated distance for the upward movement in the first iteration is **3.98 m**. The same procedure is then repeated for the remaining three iterations (as listed in Table 5), and the average of the obtained distances is computed. Based on this average value, the height between the second and third floors of the T-Building at Hochschule Bremerhaven is estimated. Figure 17 presents the acceleration, corrected velocity, and distance corresponding to the **downward movement** of the first iteration.

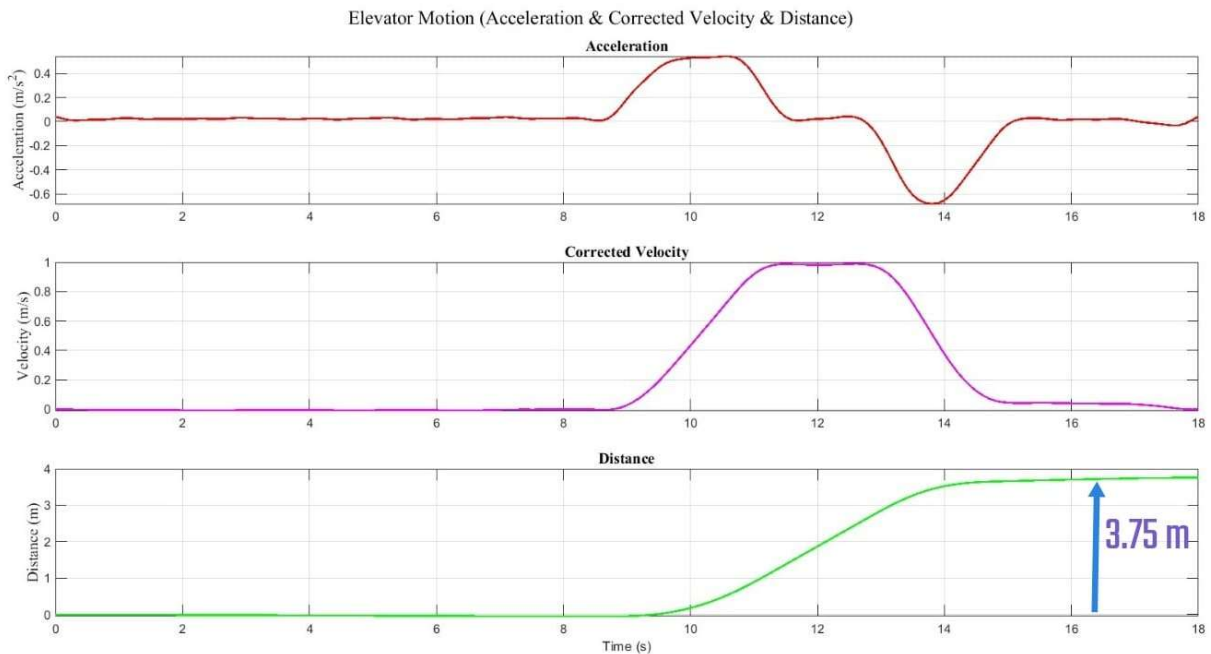


Figure 17: Acceleration, velocity, distance of downward movement of first iteration

Table 5: Distance derived from all three iterations

Iteration	Distance in upward motion (in m)	Distance in downward motion (in m)
1	3.98	3.75
2	3.54	4.03
3	3.89	4.48

$$\text{Height} = \frac{(3.98 \text{ m} + 3.75 \text{ m} + 3.54 \text{ m} + 4.03 \text{ m} + 3.89 \text{ m} + 4.48 \text{ m})}{6} = \mathbf{3.94 \text{ m}}$$

From all the calculations performed, it can be estimated and concluded that the height between the second and third floors of the T-Building at Hochschule Bremerhaven is approximately **3.94 m**.

7. Conclusion

This report presented the development and evaluation of an acceleration-based measurement system for estimating the height between elevator floors. Acceleration data recorded during multiple elevator trips were processed in MATLAB using offset correction, low-pass filtering, and drift compensation techniques. The displacement was calculated by applying discrete summation to the filtered acceleration data to obtain velocity, followed by a second summation to determine distance.

Based on the processed data, the height between the second and third floors of the T-Building at Hochschule Bremerhaven was estimated to be approximately **3.94 m**, obtained by averaging the results from multiple iterations. The result is physically reasonable; however, measurement accuracy was influenced by factors such as sensor noise, residual bias, calibration limitations, and drift accumulation caused by repeated summation.

To improve future experiments, more accurate sensor calibration, higher sampling rates, better alignment of the sensor axis, and advanced drift compensation methods are recommended. Despite existing limitations, the experiment successfully demonstrates that elevator floor height can be estimated using acceleration measurements and discrete summation methods.

8. References

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